Problem 1) $T = \frac{1}{2} m\dot{x}^2 = \frac{1}{2} m\omega_0^2 A^2 \cos^2(\omega t - \delta) = \frac{1}{2} kA^2 \cos^2(\omega_0 t - \delta)$

The potential energy gain is the same as the amount of work done by the force:

$$-dw = -F\,dx = k\,dx = dU$$

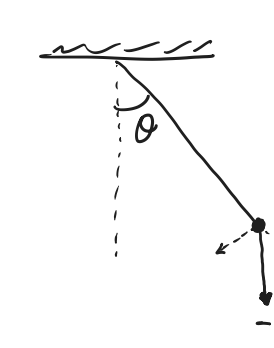
$\Rightarrow U = \frac{1}{2} kx^2 = \frac{1}{2} kA^2 \sin^2(\omega_0 t - \delta)$

$\Rightarrow E = T + U = \frac{1}{2} kA^2 \left[\cos^2(\omega_0 t - \delta) + \sin^2(\omega_0 t - \delta)\right]$

$\Rightarrow \boxed{E = \frac{1}{2} mA^2}$

Problem 2) The tangential component of $-mg$ is,

$-mg \sin\theta = m(l\ddot{\theta})$



This eqn. is not analytically solvable.
We must make a small angle approximation.

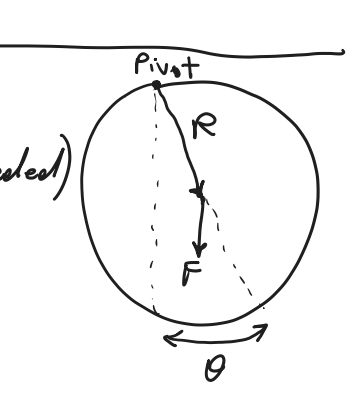
$\lim_{\theta \to 0} \sin\theta = \theta$

$\Rightarrow \ddot{\theta} + \omega^2 \theta \cong 0 \;;\; \theta \to 0 \quad$ where $\omega = \sqrt{\frac{g}{l}}$

$\Rightarrow \theta(t) \cong A\cos(\omega t + \varphi)$ for small θ.

A and φ are unknown constants determined by the initial conditions of the problem.

Problem 3) Let the rotational inertia of the sphere be I. (In reality $I = \frac{7}{5} mR^2$, but it is not needed)



The torque on the sphere is $N = R \times F$

$|N| = RF\sin\theta = Rmg\sin\theta \simeq Rmg\theta$ for $\theta \to 0$

Thus, the eqn. of motion is, $I\ddot{\theta} = -Rmg\,\theta$ (because for increasing θ, $\ddot{\theta} < 0$)

$\Rightarrow I\ddot{\theta} + Rmg\theta = 0 \Rightarrow \omega = \sqrt{\frac{Rmg}{I}} \Rightarrow \tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{Rmg}}$

Problem 4) No. If so, for the same position and velocity, two future would be possible, which is impossible for deterministic systems.

Problem 5) For the simple harmonic motion:

$x(t) = A\sin(\omega_0 t - \delta) \;;\; \dot{x}(t) = A\omega_0 \cos(\omega_0 t - \delta)$

$\Rightarrow \frac{x}{A} = \sin(\omega_0 t - \delta) \;;\; \frac{\dot{x}}{A\omega_0} = \cos(\omega_0 t - \delta)$

$\Rightarrow \left(\frac{x}{A}\right)^2 + \left(\frac{\dot{x}}{A\omega_0}\right)^2 = 1$ This is the eqn of an ellipse whose semi axes are: $A = \sqrt{\frac{2E}{k}}$ and $A\omega_0 = \sqrt{\frac{2E}{m}}$

Problem 6) A damped oscillator is described by $x(t) = e^{-\frac{\beta t}{2}} \cos(\omega_1 t - \delta)$

$\Rightarrow \dot{x}(t) = -\frac{1}{2} A e^{-\frac{\beta t}{2}} \left[\beta\cos(\omega_1 t - \delta) + 2\omega_1 \sin(\omega_1 t - \delta)\right] \;;\; \omega_0 = \sqrt{\frac{k}{m}}$

$\Rightarrow U = \frac{1}{2} kx^2 = \frac{1}{2} mA^2 \omega_0^2 e^{-\beta t} \cos^2(\omega_1 t - \delta)$

$T = \frac{1}{2} m\dot{x}^2 = \frac{1}{8} mA^2 e^{-\beta t} \left[\beta\cos(\omega_1 t - \delta) + \omega_1 \sin(\omega_1 t - \delta)\right]^2$

$E = T + U$

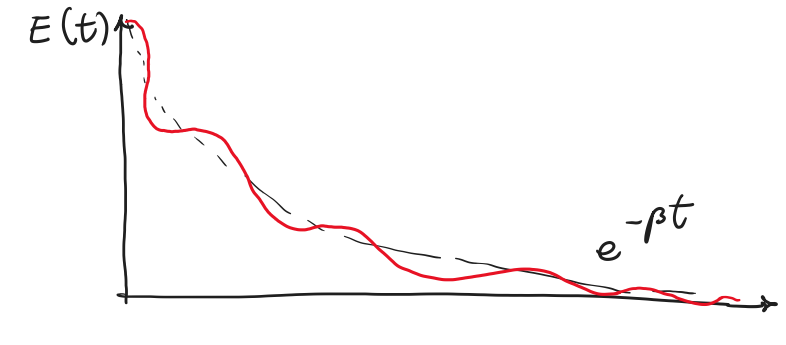
$= \frac{1}{8} mA^2 e^{-\beta t} \left(\left[\beta\cos(\omega_1 t - \delta) + 2\omega_1 \sin(\omega_1 - \delta)\right]^2 + 4\omega_0^2 \cos^2(\omega_1 t - \delta)\right)$

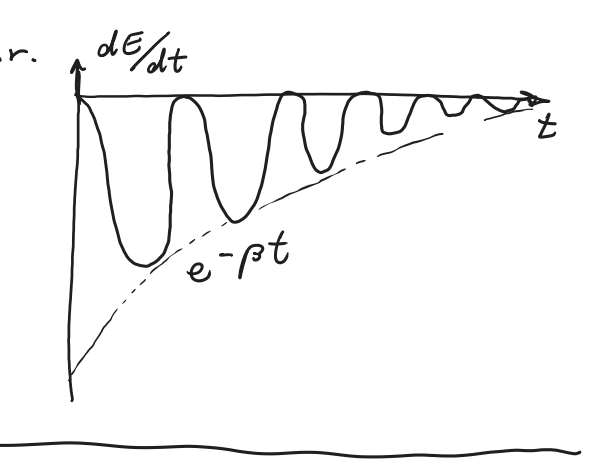
$= \frac{1}{8} mA^2 e^{-\beta t} \left(4\omega_0^2 + 2\beta\omega_1 \sin(2\omega_1 t - 2\delta) + \beta^2 \cos(2\omega_1 t - 2\delta)\right)$

$= \frac{1}{2} mA^2 \omega_0^2 e^{-\beta t} \left[1 + \frac{\beta}{2\omega_0} \cos(2\omega_1 t - \delta')\right]$ with $\delta' = \tan^{-1}\left(\frac{2\omega_1}{\beta}\right)$

It is obvious that E is modulated by the exponential $e^{-\beta t}$, which is decaying over time.

Note that when $\beta \to 0$ (no damping) $E \to \frac{1}{2} mA^2 \omega_0^2$ which is the energy of a simple harmonic oscillator.





Problem 7) Recall again the eqns for an underdamped motion:

$x(t) = Ae^{-\beta t} \cos(\omega_1 - \delta) \qquad$ where $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$

$\dot{x}(t) = -Ae^{-\beta t} \left[\beta\cos(\omega_1 t - \delta) + \omega_1 \sin(\omega_1 t - \delta)\right]$

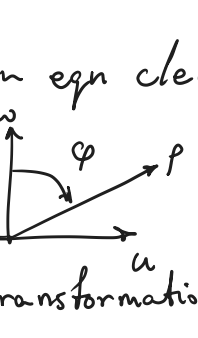
These equations are hard to digest. But we can make them more recognizable by a change of variables:

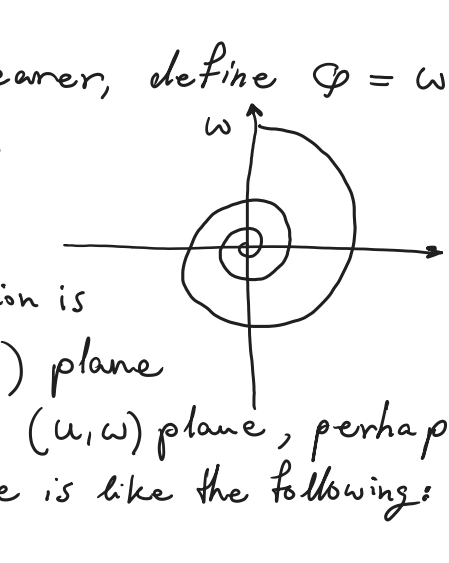
$\begin{cases} u = \omega_1 x \\ w = \beta x + \dot{x} \end{cases}$ How did we guess this?! By noting that $\dot{u} = w$ now in a simple spiral form.

$\Rightarrow \begin{cases} u = \omega_1 A e^{-\beta t} \cos(\omega_1 t - \delta) \\ w = -\omega_1 A e^{-\beta t} \sin(\omega_1 t - \delta) \end{cases} \Rightarrow u^2 + w^2 = \rho^2 = \omega_1^2 A^2 e^{-2\beta t}$

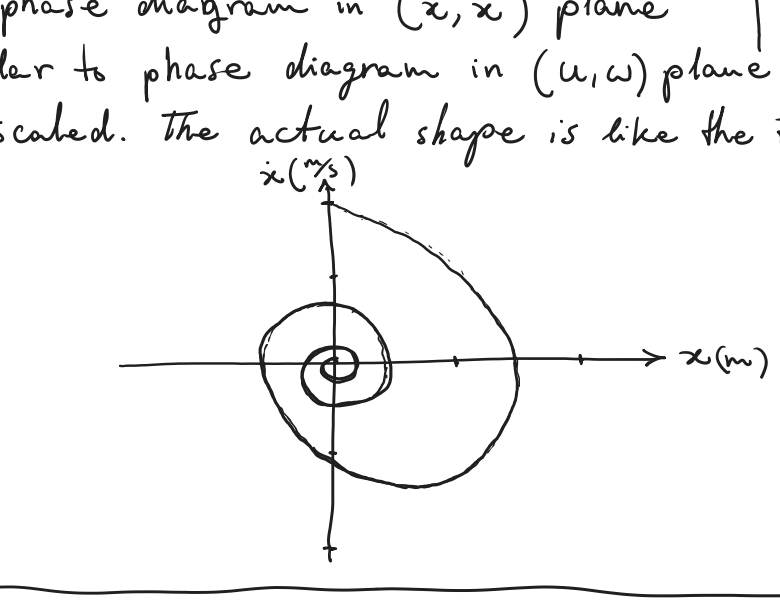
To make the inward spiral motion eqn clearer, define $\varphi = \omega_1 t$

$\Rightarrow \rho = \omega_1 A e^{-\left(\frac{\beta}{\omega_1}\right)\varphi}$





Because $(x, \dot{x}) \to (u, w)$ transformation is linear, the phase diagram in $(x, \dot{x})$ plane will look similar to phase diagram in (u, w) plane, perhaps only skewed or scaled. The actual shape is like the following:



Problem 8) Here gravity is the restoring force.

$F = m(l\ddot{\theta}) = $ Restoring + resistive force.

$ml\ddot{\theta} = -mg\sin\theta - 2m\sqrt{\frac{g}{\ell}}\,(l\dot{\theta})$; Assuming small oscillations, $\sin\theta \simeq \theta$

$\Rightarrow \ddot{\theta} + 2\sqrt{\frac{g}{\ell}}\,\dot{\theta} + \frac{g}{\ell}\theta = 0 \Rightarrow \omega_0^2 = \frac{g}{\ell} \;;\; \beta = \frac{g}{\ell} = \omega_0^2$

Thus, the pendulum is critically damped, which has the solution,

$x(t) = (A + Bt)e^{-\beta t}$; We use the initial condition,

$\theta(t=0) = \alpha$ and $\dot{\theta}(t=0) = 0$ to determine A & B.

$\dot{\theta}(t) = Be^{-\beta t} - \beta(A + Bt)e^{-\beta t} \Rightarrow \begin{cases} \theta(t=0) = \alpha = A \\ \dot{\theta}(t=0) = 0 = B - \beta A \Rightarrow B = \alpha\beta \end{cases}$

$\Rightarrow \begin{cases} \theta(t) = \alpha\left(1 + \sqrt{\frac{g}{\ell}}\,t\right) e^{-\sqrt{\frac{g}{\ell}}\,t} \\ \dot{\theta}(t) = -\frac{\alpha g}{\ell}\, t e^{-\sqrt{g/\ell}\,t} \end{cases}$

Problem 9) The complementary solution $x_c(t)$ is the same as before for the damped oscillation.

$$x(t) = e^{-\beta t}\left[A_1 \exp\left(\sqrt{\beta^2 - \omega_0^2}\,t\right) + A_2 \exp\left(-\sqrt{\beta^2 - \omega_0^2}\,t\right)\right]$$

For the particular solution, we can try,

$$x_p(t) = D\cos(\omega t - \delta)$$

Substituting $x_p(t)$ in the motion eqn; $\ddot{x} + 2\beta\dot{x} + \omega_0^2 = A\cos\omega t$

leads to, $\quad \left\{A - D\left[(\omega_0^2 - \omega^2)\cos\delta + 2\omega\beta\sin\delta\right]\right\}\cos\omega t$
$\qquad - \left\{D\left[(\omega_0^2 - \omega^2)\sin\delta - 2\omega\beta\cos\delta\right]\right\}\sin\omega t = 0$

Because sin and cos are linearly independent functions, the coefficient of each must be zero. For the $\sin\omega t$ term:

$$\tan\delta = \frac{2\omega\beta}{\omega_0^2 - \omega^2}$$

$\Rightarrow \begin{cases} \sin\delta = \dfrac{2\omega\beta}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \\ \cos\delta = \dfrac{\omega_0^2 - \omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \end{cases}$

For the $\cos\omega t$ term, we have

$$D = \frac{A}{(\omega_0^2 - \omega^2)\cos\delta + 2\omega\beta\sin\delta}$$

$$= \frac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}}$$

$\Rightarrow x_p(t) = \dfrac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \cos(\omega t - \delta)$ with $\delta = \tan^{-1}\left(\frac{2\omega\beta}{\omega_0^2 - \omega^2}\right)$

where δ represents the phase difference between the driving force and the resultant motion. The general solution is

$$x(t) = x_c(t) + x_p(t)$$

where $x_c(t)$ represents the transient effects (that die out because of its $e^{-\beta t}$ factor) and $x_p(t)$ represents the steady-state effects, $x\left(t \gg \frac{1}{\beta}\right) = x_p(t)$

The transient effect although not permanent is very important in electrical circuits.

Problem 10) We would like to know the kinetic energy resonance, that is, the frequency ω at which T is maximum.

$T = \frac{1}{2} m\dot{x}^2 \;;\; x_p(t) = \dfrac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}} \cos(\omega t - \delta)$

Although the complete solution is the sum of x_c and x_p, we can safely ignore $x_c(t)$ because it is only a transitive effect that will vanish over time.

$\Rightarrow \dot{x} = \dot{x}_p(t) = \dfrac{-A\omega}{\sqrt{(\omega^2 - \omega_0^2)^2 + 4\omega^2\beta^2}} \sin(\omega t - \delta)$

$\Rightarrow T = \dfrac{mA^2}{2} \dfrac{\omega^2}{(\omega^2 - \omega_0^2)^2 + 4\omega^2\beta^2} \sin^2(\omega t - \delta)$

This T varies over time. Thus, we need to compute the average T over one period to find ω_E.

$\Rightarrow \langle T \rangle = \dfrac{mA^2}{2} \dfrac{\omega^2}{(\omega^2 - \omega_0^2) + 4\omega^2\beta^2} \langle \sin^2(\omega t - \delta) \rangle$

$\langle \sin^2(\omega t - \delta) \rangle = \dfrac{\omega}{2\pi} \displaystyle\int_0^{\frac{2\pi}{\omega}} \sin^2(\omega t - \delta)\,dt = \frac{1}{2}$

$\Rightarrow \langle T \rangle = \dfrac{MA^2}{4} \dfrac{\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}$

$\dfrac{d\langle T \rangle}{d\omega} = 0 \Rightarrow \boxed{\omega_E = \omega_0}$ which is different from the amplitude resonance frequency $\omega_A = \sqrt{\omega_0^2 - 2\beta^2}$.

Because the potential energy directly depends on the amplitude, the potential energy resonance frequency must also occur at ω_A.

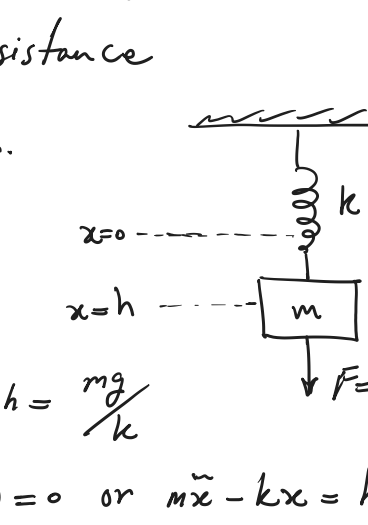
The fact that resonances occur at different frequencies is not surprising because the system is constantly dissipating energy into the medium.

Problem 11) Mechanical Force $(mg) \equiv$ Electromagnetic force (emf)

The damping parameter $b \equiv$ electrical resistance

The displacement $x \equiv$ electrical charge q.

$\frac{1}{k} \equiv C,\; m \to L,\; F \to \mathcal{E},\; \dot{x} \to I$



The equilibrium point x for the mass m can be obtained as, $\Sigma F = 0 = mg - kh \Rightarrow h = \frac{mg}{k}$

Thus, the equation of motion is $m\ddot{x} + k(x - h) = 0$ or $m\ddot{x} - kx = kh$ whose solution is $x(t) = h + A\cos\omega_0 t$ with $\begin{cases} x(t=0) = h + A \\ \dot{x}(t=0) = 0 \end{cases}$

The equivalent electrical circuit becomes,

Ⓘ $L\frac{dI}{dt} + \frac{1}{C}\int I\,dt = \mathcal{E} = \frac{q_c}{C}$ with q_c representing the charge that must be applied to the capacitor to produce a voltage $\mathcal{E}$. Since $I \equiv \dot{q} \Rightarrow L\ddot{q} + \frac{q}{C} = \frac{q_c}{C}$

For $q(t=0) = q_0$ and $\dot{q}(t=0) = 0$ the solution becomes:

$q(t) = q_c + (q_0 - q_c)\cos\omega_0 t$

Note that we used Kirchhoff's loop rule to derive eqn Ⓘ:

$\Sigma V_i = 0$. For the solenoid $V_L(t) = L\frac{dI}{dt} = L\ddot{q}$

For the capacitor, $C = \frac{dQ}{dV} \Rightarrow dV_C(t) = \frac{dQ}{C} = \frac{q(t)}{C}$

$V_L + V_C = \mathcal{E} = \frac{q_c}{C}$

Problem 12) $V(x) = V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2!}V''(x_0)(x - x_0)^2$
$\qquad + \frac{1}{3!}V'''(x_0)(x - x_0)^3 + \cdots$

For equilibrium, $V(x_0) = 0 \Rightarrow V(x) \simeq \frac{1}{2}V''(x_0)(x - x_0)^2$

This is the Hooke's law! $F = -\frac{dV}{dx}$ and we know $\omega = \sqrt{\frac{V''(x_0)}{m}}$

With this in mind, Let's check the potential $V(x) = \frac{A}{x^2} - \frac{B}{x}$

with $A, B > 0 \Rightarrow V'(x) = -\frac{2A}{x^3} + \frac{B}{x^2}$; An equilibrium point requires $V'(x) = 0 \Rightarrow \boxed{x_0 = \frac{2A}{B}}$ the equilibrium point.

$V''(x) = \frac{64}{x^4} - \frac{2B}{x^3}\Big|_{x = x_0} = \frac{2A}{B} \Rightarrow \omega = \sqrt{\frac{V''(x_0)}{m}} = \sqrt{\frac{B^4}{8MA^3}}$

Problem 13) $L = T - U$; The kinetic energy is composed of radial and tangential components,

$$T = \frac{1}{2}m\left(\dot{x}^2 + \underbrace{\left((\ell + x)\dot{\theta}\right)^2}_{\text{tangential speed}}\right) \quad \text{Ⓘ}$$

The potential energy has gravity and spring components;

$$U(x, \theta) = -mg(\ell + x)\cos\theta + \frac{1}{2}kx^2 \quad \text{Ⓘ Ⓘ}$$

Ⓘ Ⓘ Ⓘ $\Rightarrow L = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) + mg(\ell + x)\cos\theta - \frac{1}{2}kx^2$

There are two variables here: (x, θ)

$\Rightarrow \begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow m\ddot{x} = m(\ell + x)\dot{\theta}^2 + mg\cos\theta - kx \quad \text{Ⓘ Ⓘ Ⓘ} \\ \frac{\partial L}{\partial \theta} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow \frac{d}{dt}\left(m(\ell + x)^2\dot{\theta}\right) = -mg(\ell + x)\sin\theta \quad \text{Ⓘ Ⓥ} \end{cases}$

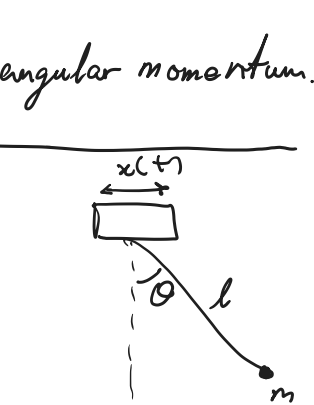
$\Rightarrow m(\ell + x)^2\ddot{\theta} + 2m(\ell + x)\dot{x}\dot{\theta} = -mg(\ell + x)\sin\theta$

$\Rightarrow m(\ell + x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg\sin\theta \quad$ Ⓥ

Ⓘ Ⓘ Ⓘ is simply the Newton's law $F = ma$ with the second term being related to the centripetal acceleration $-(\ell + x)\dot{\theta}^2$.

Ⓘ Ⓥ states that torque equals the rate of change of angular momentum.

Problem 14) $x(t) = A\cos(\omega t)$ Ⓘ



$\Rightarrow (X, Y)_m = (x + l\sin\theta, -l\cos\theta)$

$\Rightarrow V_m^2 = \dot{X}^2 + \dot{Y}^2 = \ell^2\dot{\theta}^2 + \dot{x}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta$

$\Rightarrow L = \frac{1}{2}m\left(\ell^2\dot{\theta}^2 + \dot{x}^2 + 2\ell\dot{x}\dot{\theta}^2\cos\theta\right) + mg\ell\cos\theta$

$\frac{\partial L}{\partial \theta} = \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} \Rightarrow \frac{d}{dt}\left(m\ell^2\dot{\theta} + m\ell\dot{x}\cos\theta\right) = -m\ell\dot{x}\dot{\theta}\sin\theta - mg\ell\sin\theta$

$\Rightarrow \ell\ddot{\theta} + \ddot{x}\cos\theta = -g\sin\theta$ Ⓘ Ⓘ

Ⓘ, Ⓘ Ⓘ $\Rightarrow \ell\ddot{\theta} - A\omega^2\cos(\omega t)\cos\theta + g\sin\theta = 0$

Assuming $\sin\theta \simeq \theta \Rightarrow \ddot{\theta} + \omega_0^2\theta = a\omega^2\cos(\omega t)$; $\omega_0 = \sqrt{\frac{g}{\ell}}$; $a = \frac{A}{\ell}$

which implies a driven oscillation where we have also used,

$\cos\theta\Big|_{\theta \to 0} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

$\Rightarrow \theta(t) = \frac{a\omega^2}{\omega_0^2 - \omega^2}\cos(\omega t) + B\cos(\omega_0 t - \varphi)$ with B and φ given by initial conditions. What happens if $\omega \to \omega_0$?

5.2 Small oscillations

Consider an object in one dimension, subject to the potential $V(x)$. Let the object initially be at rest at a local minimum of $V(x)$, and then let it be given a small kick so that it moves back and forth around the equilibrium point. What can we say about this motion? Is it simple harmonic? Does the frequency depend on the amplitude?

It turns out that for small amplitudes, the motion is indeed simple harmonic, and the frequency can easily be found, given $V(x)$. To see this, expand $V(x)$ in a Taylor series around the equilibrium point, x_0,

$$V(x) = V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2!}V''(x_0)(x - x_0)^2 + \frac{1}{3!}V'''(x_0)(x - x_0)^3 + \cdots. \quad (5.18)$$

This looks like a bit of a mess, but we can simplify it greatly. $V(x_0)$ is an irrelevant additive constant. We can ignore it because only differences in energy matter (or equivalently, because $F = -dV/dx$). And $V'(x_0) = 0$, by definition of the equilibrium point. So that leaves us with the $V''(x_0)$ and higher-order terms. But for sufficiently small displacements, these higher-order terms are negligible compared with the $V''(x_0)$ term, because they are suppressed by additional powers of $(x - x_0)$. So we are left with[5]

$$V(x) \approx \frac{1}{2}V''(x_0)(x - x_0)^2. \quad (5.19)$$

But this has exactly the same form as the Hooke's-law potential, $V(x) = (1/2)k(x - x_0)^2$, provided that we let $V''(x_0)$ be our "spring constant" k. Equivalently, the force is $F = -dV/dx = -V''(x_0)(x - x_0) \equiv -k(x - x_0)$. The frequency of small oscillations, $\omega = \sqrt{k/m}$, is therefore

$$\omega = \sqrt{\frac{V''(x_0)}{m}}. \quad (5.20)$$

---

[5] Even if $V'''(x_0)$ is much larger than $V''(x_0)$, we can always pick $(x - x_0)$ small enough so that the third-order term is negligible. The one case where this is not true is when $V''(x_0) = 0$. But the result in Eq. (5.20) is still valid in this case. The frequency ω just happens to be zero, in the limit of infinitesimal oscillations.

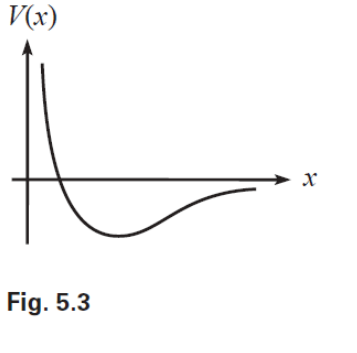


Fig. 5.3

Example: A particle moves under the influence of the potential $V(x) = A/x^2 - B/x$, where $A, B > 0$. Find the frequency of small oscillations around the equilibrium point. This potential is relevant to planetary motion, as we'll see in Chapter 7. The rough shape is shown in Fig. 5.3.

Solution: The first thing we need to do is calculate the equilibrium point x_0. The minimum occurs where

$$0 = V'(x) = -\frac{2A}{x^3} + \frac{B}{x^2} \implies x = \frac{2A}{B} \equiv x_0. \quad (5.21)$$

The second derivative of $V(x)$ is

$$V''(x) = \frac{6A}{x^4} - \frac{2B}{x^3}. \quad (5.22)$$

Plugging in $x_0 = 2A/B$, we find

$$\omega = \sqrt{\frac{V''(x_0)}{m}} = \sqrt{\frac{B^4}{8mA^3}}. \quad (5.23)$$

parabola

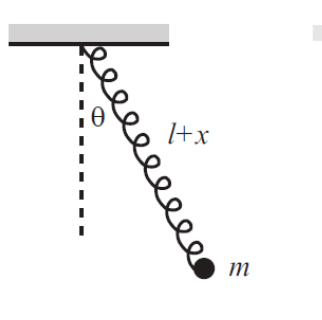


Fig. 6.1

Example (Spring pendulum): Consider a pendulum made of a spring with a mass m on the end (see Fig. 6.1). The spring is arranged to lie in a straight line (which we can arrange by, say, wrapping the spring around a rigid massless rod). The equilibrium length of the spring is ℓ. Let the spring have length $\ell + x(t)$, and let its angle with the vertical be $\theta(t)$. Assuming that the motion takes place in a vertical plane, find the equations of motion for x and θ.

Solution: The kinetic energy may be broken up into the radial and tangential parts, so we have

$$T = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right). \quad (6.9)$$

The potential energy comes from both gravity and the spring, so we have

$$V(x, \theta) = -mg(\ell + x)\cos\theta + \frac{1}{2}kx^2. \quad (6.10)$$

The Lagrangian is therefore

$$L \equiv T - V = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) + mg(\ell + x)\cos\theta - \frac{1}{2}kx^2. \quad (6.11)$$

There are two variables here, x and θ. As mentioned above, the nice thing about the Lagrangian method is that we can just use Eq. (6.3) twice, once with x and once with θ. So the two Euler–Lagrange equations are

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x} \implies m\ddot{x} = m(\ell + x)\dot{\theta}^2 + mg\cos\theta - kx, \quad (6.12)$$

and

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \frac{\partial L}{\partial \theta} \implies \frac{d}{dt}\left(m(\ell + x)^2\dot{\theta}\right) = -mg(\ell + x)\sin\theta$$
$$\implies m(\ell + x)^2\ddot{\theta} + 2m(\ell + x)\dot{x}\dot{\theta} = -mg(\ell + x)\sin\theta$$
$$\implies m(\ell + x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg\sin\theta. \quad (6.13)$$

Equation (6.12) is simply the radial $F = ma$ equation, complete with the centripetal acceleration, $-(\ell + x)\dot{\theta}^2$. And the first line of Eq. (6.13) is the statement that the torque equals the rate of change of the angular momentum (this is one of the subjects of Chapter 8). Alternatively, if you want to work in a rotating reference frame, then Eq. (6.12) is the radial $F = ma$ equation, complete with the centrifugal force, $m(\ell + x)\dot{\theta}^2$. And the third line of Eq. (6.13) is the tangential $F = ma$ equation, complete with the Coriolis force, $-2m\dot{x}\dot{\theta}$. But never mind about this now. We'll deal with rotating frames in Chapter 10.[2]

---

[2] Throughout this chapter, I'll occasionally point out torques, angular momenta, centrifugal forces, and other such things when they pop up in equations of motion, even though we haven't covered

REMARK: After writing down the E–L equations, it is always best to double-check them by trying to identify them as $F = ma$ and/or $\tau = dL/dt$ equations (once we learn about that). Sometimes, however, this identification isn't obvious. And for the times when everything is clear (that is, when you look at the E–L equations and say, "Oh, of course!"), it is usually clear only *after* you've derived the equations. In general, the safest method for solving a problem is to use the Lagrangian method and then double-check things with $F = ma$ and/or $\tau = dL/dt$ if you can. ♣

6.3. Pendulum with an oscillating support

Let θ be defined as in Fig. 6.39. With $x(t) = A\cos(\omega t)$, the position of the mass m is given by

$$(X, Y)_m = (x + \ell\sin\theta, -\ell\cos\theta). \quad (6.107)$$

Taking the derivative to obtain the velocity, we find that the square of the speed is

$$V_m^2 = \dot{X}^2 + \dot{Y}^2 = \ell^2\dot{\theta}^2 + \dot{x}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta, \quad (6.108)$$

which also follows from applying the law of cosines to the horizontal $\dot{x}$ and tangential $\ell\dot{\theta}$ parts of the velocity vector. The Lagrangian is therefore

$$L = \frac{1}{2}m(\ell^2\dot{\theta}^2 + \dot{x}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta) + mg\ell\cos\theta. \quad (6.109)$$

The equation of motion for θ is

$$\frac{d}{dt}(m\ell^2\dot{\theta} + m\ell\dot{x}\cos\theta) = -m\ell\dot{x}\dot{\theta}\sin\theta - mg\ell\sin\theta$$
$$\implies \ell\ddot{\theta} + \ddot{x}\cos\theta = -g\sin\theta. \quad (6.110)$$

Plugging in the explicit form of $x(t)$, we have

$$\ell\ddot{\theta} - A\omega^2\cos(\omega t)\cos\theta + g\sin\theta = 0. \quad (6.111)$$

In retrospect, this makes sense. Someone in the reference frame of the support, which has horizontal acceleration $\ddot{x} = -A\omega^2\cos(\omega t)$, may as well be living in a world where the acceleration from gravity has a component g downward and a component $A\omega^2\cos(\omega t)$ to the right. Equation (6.111) is just the $F = ma$ equation in the tangential direction in this accelerating world.

A small-angle approximation in Eq. (6.111) gives

$$\ddot{\theta} + \omega_0^2\theta = a\omega^2\cos(\omega t), \quad (6.112)$$

where $\omega_0 \equiv \sqrt{g/\ell}$ and $a \equiv A/\ell$. This equation is simply the equation for a driven oscillator, which we solved in Chapter 4. The solution is

$$\theta(t) = \frac{a\omega^2}{\omega_0^2 - \omega^2}\cos(\omega t) + C\cos(\omega_0 t + \phi), \quad (6.113)$$

where C and ϕ are determined by the initial conditions.

If ω happens to equal ω_0, then it appears that the amplitude goes to infinity. However, as soon as the amplitude becomes large, our small-angle approximation breaks down, and Eqs. (6.112) and (6.113) are no longer valid.